2018

COMPUTER SCIENCE – GENERAL

Second Paper

(Formal Languages and Automata Theory)

8. (a) Design a "mod 2 counter" i.e., an FSM whose input sequence is a sequence $(x_1, x_2, \ldots x_n)$ of 0_s and 1_s, and whose output sequence $(y_1, y_2, \ldots y_n)$ indicates the arrival of every other input 1, i.e,

$$y_n \begin{cases} 1 \text{ if } x_n = 1 \text{ and if there have been an even number of input 1s so far.} \\ 0 \text{ in all other cases.} \end{cases}$$

(b) Write a brief note on Chomsky classification of grammar.

(c) Find a grammar generating $\left\{ a^j\, b^n\, c^n \,\middle|\, n \geq 1,\ j \geq 0 \right\}$ 4+6+6

9. Consider the following grammar :

(a) (S is the starting symbol, and other symbols have their usual meaning)

$S \rightarrow AB;\ A \rightarrow A \cdot 1 | 0;\ B \rightarrow 2B | 3$. Identify the grammar according to Chomsky classification of grammar.

(b) State the Pumping Lemma for context free languages. What is its significance?

(c) Consider a Mealy machine given below. Construct a Moore machine equivalent to this Mealy machine. 6+(4+2)+4